# LESSON PLAN

## SFNA (MOD AIRMANSHIP) (24)

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| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 02 |
| **Total No. of Lesson Plans** | 6 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

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| **1. Topic** | Introduction to lubricating oils and high-grade oils, nature of oils, oil contamination |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Introduction to lubricating oils and high-grade oils, nature of oils, oil contamination |
| **5. Specific Objectives** | 24.1.1.1 Trainee will be able to describe the Various nomenclatures |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

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| **7. Practical in Lab/Equipment Rooms/Simulators** | Demonstration of oil sample identification and contamination visual inspection; Safety: Wear gloves and eye protection; Supervision: Instructor monitors handling of samples |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

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| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What is the primary function of lubricating oil in an aircraft engine?
(b) Name one common source of oil contamination in aviation systems.
(c) Why is understanding oil nomenclature important for maintenance personnel?

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Definition and classification of lubricating oils<br>• Distinction between high-grade and standard oils<br>• Chemical nature and composition of base oils<br>• Role of additives in oil formulation<br>• Standard nomenclature codes for aviation oils | L & Ds | PPT, IB | 15 min |
| (b) | • Sources and types of oil contamination<br>• Impact of contamination on engine performance<br>• Visual identification of common contaminants<br>• Sampling procedures for contamination detection<br>• Preventive measures against oil degradation | L & Ds | PPT, PDF | 20 min |
| (c) | • Review of key nomenclature terms<br>• Case study on oil failure due to contamination<br>• Q&A session on oil properties and usage<br>• Clarification of complex terminology<br>• Linking nomenclature to maintenance procedures | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain the difference between high-grade and standard lubricating oils based on their nomenclature.
(b) Describe two major consequences of oil contamination on aircraft engine operation.
(c) How does the nomenclature system help in selecting the correct oil for specific engine types?

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| **12. Home Assignment and Follow Up** | Prepare a brief report comparing the nomenclature and typical applications of two different high-grade aviation oils discussed in this lesson. |

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| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | Various properties of FOLG's viscosity, absolute kinematics and dynamic viscosity, viscosity index, extreme pressure lubricants, film lubrication and boundary lubrication, determination of viscosity and viscosity index, activities in oils, service ratings of oil, SAE number | Refer to the master syllabus reference list | LP 03 |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
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**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min